Tissue culture

Tissue Culture

- Remove (isolation) of cells from animal or plant and their subsequent growth in favorable Artificial media under control conditions

- **Different types of cells grown in culture :**

- Connective tissue cells

Fibroblasts, skeletal cells, cardiac cell.

- Epithelial cells

Liver cells, breast, skin, kidney.

- Tumor cells.

Primary culture

- Culture contain cells directly isolate from parental tissue.
- Primary cells Proliferate under the appropriate conditions until they occupy all Available substrate.
- Substrate→ surface where cells attach.
- The mostb cells require attach to surface of plate or certain substance like collagen to adapt with new environment (tissue is the origin environment), transfer cells from environment to new environment make the cells need to Adapt with new environment, adaptive essential to cell growth (divide).
- Some cells require attach to substrate, other don’t, depend on the cell type.

- Most cells in the tissues are attached, to isolate these cells · using enzymes mainly trypsin or using mechanical means.

- Primary culture contain cells directly isolate from animal or plant tissue which mean, culture contain very Heterogeneous population of cell (many different types of cells not only one type).

- In tissue culturing, at certain stage cells must be subculture to new flask or plate with fresh growth media, because at certain point when the cells occupy 100% of plate, growth stop and high cells die.

- Before cells reach 100% of plate surface cell must passage (subculture).

- Culture produced After subculture of primary culture · secondary culture.

1-Primary culture : contain cells directly isolate from parental tissue (host tissue).

2-Secondary culture : contain cells from primary culture.

- Lifetime of cells in primary culture

- Slow growth in the beginning
- Exponential growth
- Cell senescence and death .

- Classically, the lifetime of cells (hayflick) approximately 50 generations.

- Which mean cell line can passage 50 times.
- Passage number : how many time cells are passaged.

Passage number = 5 → cells are good for using in experiments

Passage number =40→ cells are bad for using in experiments.

- Hayflick limit depend on :

1.Telomere length

2.Telomeres activity.

3.DNA repair machinery.

4.Other factors as variables that contribute to senescence.

Cell lines

- Generate after subculture of primary culture.

- Two types of cell lines

1. Finite

2. Infinite

- Finite cell line : have limit life span, they Passage several times before they losing ability of Proliferate “Senescent”.

- Most types of primary cells used in research (Finite)

- Epithelial cells
- Fibroblasts
- Keratinocytes
- Melanocytes
- Endothelial cells
- Muscle cells
- Hematopoietic stem cells
- Mesenchymal stem cells

- Culture only remain for limited period of time

Infinite or continuous (immobilized) cell line

- Some cell line become immortal (divide indefinitely) through process called transformation.

- Transformation occur by :

1. Spontaneous
2. Chemicals
3. Viruses

- When Finite cell lines undergo transformation convert to Infinite cell lines which have ability to divide indefinitely.

- Primary culture of finite cell lines prepare from normal cells.
- Primary culture of Infinite (continuous) cell lines prepare from tumors.
- Some tumor cell attach within tissue other not attach.
- If the cells attach, use Trypsin or mechanical means to dissociated cells from tissue.
- Myloma or other blood cells cancers not attach can isolate without trypsin

Characteristics of Infinite cell lines

- Short generation time (12-24 h)
- Have aneuploidy chromosomes number
- Anchorage independent→ divide and function in the Absence of substratum (Attachment not essential)
- Less adherent
- Grow in Suspension
- Have tendency to grow up to each other (multilayers)
- Different in phenotype from donor tissue.

Transformation vs Transfection

- Transformation : Spontaneous or induced permanent phenotypic changes resulting from change in DNA and Gene expression.

- Transformation Effects :

- Growth rate
- Mode of growth (loss of contact inhibition)

- Normally when cells growth and reach certain point in which the number of cells increase until the cells are touch each other (crowding) the cells stop divide this called contact inhibition.

- Transformed cells don’t undergo contact inhibition.

- Specialized product formation

- Longevity

- Loss need of Adhesion.

- Transfection :introduce DNA into a cells.

Transformation

- Associated with genetic instability and may lead to malignancy.

- Variability among cancer cells in phenotype and genotype in single tumor due interaction with other cells and change in microenvironment.

- Immortal cell lines are a very important tool for research.

- Stem cells are completely different from cancer cells, both have ability to divide indefinitely, but stem cells divide then differentiate into normal part of body, while cancer cells divide and produce Abnormal tumor.

- Regardless the ability of stem cells to continuous divide, can undergo morphological, phenotypical, and genetic changes.

- Media composition can control these changes.

Continuous cell lines (Infinite)

- Permanently established cell culture that will proliferate indefinitely given appropriate fresh medium and space.

- As they are sub cultured , they become different from the original cell following interaction with different cell environment : cancer cell lines.

- more robust and easier to work with than primary cells. They have unlimited growth potential and are a quick, easy way to get basic information.

Continuous cell line are

1. Easier to maintain in vitro,
2. have faster and longer growth
3. less nutritional requirement
4. high plating efficiency.
5. they lead to highly reproducible results
6. They have unlimited growth potential.

- Drawbacks to working with continuous cell line

- Continuous cell lines are Transformed (genetically modified which can alter physiological properties also extensive Passage cause changes.

Adherent vs Suspension culture

- Include→ fibroblasts and epithelial cells

Adherent cells

- Cells need substratum Anchorage dependent substratum is prerequisite, before culturing the cells, substratum must exist.
- Adherent cells subjected to contact inhibition, grow and divide until the cells touch each other.
- Adherent cells grow as mono layer.
- Using Trypsin to dissociate cells from parental tissue, because the cells are attached.

- Using phosphate Buffer saline (PBS) for washing dissociated cells.

- PBS used for washing Adherent cells (free of Ca and Mg), because the These ions essential for Attachment.

- For Passaging (subculture) of Adherent cells, using trypsin/EDTA cover mono layer cells, then incubate for 1-2min at 37c°.

- After transfer the cells to new bottle / flask contain fresh media, cells start reattach.

Suspension cells

- Anchorage independent, don’t attach to surface of culture vessels
- Free floating not Adherent
- Using Trypsin is not requirement
- Grow much faster
- Have short lag period (quickly adapt to new environment).
- Easily maintain
- Do not require the frequent replacement of the medium.
- They are Homogeneous type.
- Include : Hematopoietic stem cells and tumor cells.

Tissue culture expression

- Confluence: percentage culture vessels surface area that appear cover by cells when observed by microscope.

-50% confluence →half of surface culture, vessels, bottle flask is covered in cells, while other 50% of the surface is not cover by cells.

-100% confluence→ surface of tissue culture completely cover by cells.

- Passage number :refers to how many times the cell has been sub-cultured.

-It gives an indication of how old the cells.

- Generation No: the number of population doublings of a cell line since isolation.

- Doubling time: the time required for a culture to double in number.

- The generation time: is the time interval required for the cells to divide.

Subculture(Passaging)

- The removal of the medium and transfer of cells from a previous culture into fresh growth medium.

- In log phase cells Proliferate exponential.

- When the cells in adherent cultures occupy all the available substrate and have no room left for expansion, or when the cells in suspension cultures exceed the capacity of the medium to support further growth, cell proliferation is greatly reduced or ceases entirely.

- To keep cells at optimal density for continuous growth cell culture has to divide and fresh media supply.

- Subculturing must be at log phase before the cells reach 100% confluence.

- Approximately at 80% confluence.

- Repeated Subculturing of cancer cells tend to transform their Characteristics.

When to Subculture?

1. **Cell density**

When the cell density (cells/cm2 substrate) reaches a level such that all of the available substrate is occupied, or when the cell concentration (cells/mL medium) exceeds the capacity of the medium, growth ceases or is greatly reduced.

- Medium must be change
- Or Culture must be divide

- Cultures at a high cell concentration exhaust the medium faster than those at a low concentration.

- Adherent cultures should be passaged when they are in the log phase, before they reach confluence(100% confluence).

- Normal cells stop growing when they reach confluence (contact inhibition), and it takes them longer to recover when reseeded (high risk of losing cells).

- Transformed cells can continue divide even after reach 100% confluence, but usually deteriorate after two doubling

- Also suspension cells must Passage before reach 100% confluence (at log phase).

- When the suspension cells reach confluence, cells clump to gather and medium become turbid.

2. **Deterioration of morphology**

- Cells must be examine by microscope, to ensure the cells have normal shape, if the shape start change , cells should be passage or media must change.

- If the deterioration is progressed too much, it become irreversible and cells will die.

- Media used in cell culturing contain PH indicator

- (phenol red) , because cells are sensitive to PH change,

- Most cells growth at pH=7.4

- Most cells stop growing when PH fall from 7 to 6.5

- Cells start loss viability when PH fall from 6.5 to 6

- Cells exhaust media and produce waste (lactic acid), Lactic Acid is byproduct of cellular metabolic process.

- Accumulation of Lactic Acid cause drop in PH, which lead to change the media from red color to orange to Yellow color.

- Cells must Passage before media color become yellow.

- Increase in cells→ increase media exhausting · increase in lactic acid accumulation.

- If the PH drop in 0.1 at day, media not necessarily change daily.

- If the PH drop 0.5 at day, media must be change as much as possible.

3.Cell type

- Normal cells (fibroblasts) usually stop dividing at a high cell density, because of cell crowding, growth factor depletion, and other reasons. The cells block in the G1 phase of the cell cycle and deteriorate very little, even if left for two to three weeks or longer.

- Transformed cells, and continuous cell lines, however, deteriorate rapidly at high cell densities unless the medium is changed daily or they are subcultured

Tissue culturing are developed due to :

- Using Anti-biotic
- Using Trypsin
- Using culture defined media

Types of cells in culture based on shape and morphology:

- Fibroblastic (or fibroblast-like) cells are bipolar or multipolar, have elongated shapes, and grow attached to a substrate.

- Epithelial-like cells are polygonal in shape with more regular dimensions, and grow attached to a substrate in discrete patches.

- Lymphoblast-like cells are spherical in shape and usually grown in suspension without attaching to a surface.

Application of cell culture :

1)Using in cellular and molecular biology

2)Study drug effect and toxicity of compounds

3)Drug screening and development

4)Virology

5)Genetic Engineering.

Tissue culture Lab

- Location : should be far from microbiology labs or animal houses
- Should not open directly to corridors with heavy pedestrian.
- All equipment, tools inside the lab restricted to lab never Barrow or allow any one use equipment for other labs (microbiology, Biochemistry, or any lab).

- Make sure that only equipment need only exist in Lab.
- To facilitate cleaning, the walls should be painted with waterproof, oil-based gloss paint.
- Primary cell line should never be kept in the same incubator or Laminar flow hood as hybridoma cells.

Basic requirements

- Cell culture hood

(i.e., laminar-flow hood or biosafety cabinet)

- Incubator (humid CO_2 incubator recommended)
- Water bath
- Centrifuge
- Refrigerator and freezer (–20°C)
- Cell counter (e.g., Countess® Automated Cell Counter or hemacytometer)
- Inverted microscope
- Liquid nitrogen (N_2) freezer or cryostorage container
- Sterilizer (i.e., autoclave)

Expanded equipment

- Aspiration pump (peristaltic or vacuum)
- PH meter
- Confocal microscope
- Flow cytometer

Additional Supplies

- Cell culture vessels (flasks, Petri dishes, roller bottles, multi-well plates(
- Pipettes and pipettors
- Syringes and needles
- Waste containers, Media, sera, and reagents Cells.

Laminar flow (cell culture hood)

- Provide an aseptic work area by reduce and minimize contamination as much as possible.

- Three classes of Cell culture hood :
- Class I
- Class II
- Class III

- **Class I**
- Significant production of person
- Significant production of environment
- Don’t provide culture production form contamination.
- Contamination not only infection (Bacteria, viruses, fungi) but also include chemicals, cells from human, if the person work without gloves it may cells from person hand contaminate the culture.

- **Class II**

- Provide protection essential for tissue culture

- Biosafety biological cabinet should be used for handling potential hazardous material (viral infected culture, toxic Reagents, radio isotopes)

- **Class III**

- Gas tight
- Provide highest protection level to personal and environment.
- Used for work with human known pathogens.

Horizontal vs vertical hoods

Some hoods→ air flow is parallel to worker (horizontal)

Others →air flow from the top to worker area (vertical).

- Horizontal flow hood :provides protection to the culture (if the air flowing towards the user) or to the user (if the air is drawn in through the front of the cabinet by negative air pressure inside).

- A Vertical flow hoods, on the other hand, provide significant protection to the user and the cell culture.

Both types contain :

- Pre-filter : removes most particulate matter.

- the high efficiency particle filters (HEPA)

capable of removing from air particulate matter that is greater than 0.3 μm in diameter. This means that most but not all Bacteria and fungi spores will be removed.

- Air flow must measure Regularly by Anemometer.

Incubator

- Provide the appropriate conditions for cell growth
- Incubator must be large enough to your laboratory needs.
- Temperature must be control (+/- 0.2).
- Made from stainless steel to easy cleaning and corrosion protection.
- Must be cleaning Regularly to prevent contamination.

Types of incubator

- **Dry incubator** :more economic, but require the cell cultures to be incubated in sealed flasks to prevent evaporation.

- Placing a water dish in incubator to provide some humidity, but not Precise control of the conditions in incubator.

- **Humid CO2 incubators** : are more expensive, but allow superior control of culture conditions.

- They can be used to incubate cells cultured in Petri dishes or multi-well plates, which require a controlled atmosphere of high humidity and increased CO_2 tension.

Refrigerator

- **Small Refrigerator**

- present in small laboratories.
- In expensive
- Storing media and reagents at 2-8 c°

- **Large Refrigerato**
- Cold room only for cell cultures
- Make sure clean of room Regularly to avoid contamination.

Centrifuge

Refrigerator bench to, rotate at rate 1500rPm or 499 g.

Inverted microscope

- For tissue culturing, special type of light microscope use “inverted microscope”, inverted microscope use light to visualize, but inverted to provide space for putting culture or flasks that containing cells.

- Cell cultures must be examine Regularly by microscope to make sure the culture is pure.

- For Hybridoma using inverted microscope

- Eye piece = 10X-15X

- Objective= 10X – 20X

Liquid nitrogen

- Minus 196c° (- 196)
- Using for cryogenic storage (long term storage)
- Liquid nitrogen essential for long term storage to protect cell from genetic instability.
- DON’T storage cells for long term at - 20 C° or – 80C° because the viability of cells quickly decrease
- Vapor phase systems minimize the risk of explosion with cryostorage tubes, and are required for storing biohazardous materials,
- liquid phase systems usually have longer static holding times, and are therefore more economical.

Cell counter

- Essential for quantitative growth kinetics, and a great advantage when more than two or three cell lines are cultured in the laboratory.

- Automated cell counter :measure cell count and viability (live, dead, and total cells) in less than minute.

- Hemacytometer :counting chamber use for manual counting.

Sterilization

- Autoclave : using instrument combine with heat, pressure, and time to kill all form of Microorganisms (bacteria, spores, viruses,…..)
- Dry heat by oven combine between heat and time.

- Complete sterilization

- 120 c°
- 10- 15 lb pressure
- 20 min

- For solutions Sterilization, bottle cap should be loosely closed, after finish the make sure close the bottle cape tightly.

- Glass Pipettes cover with cotton wool and autoclaved in sealable metal container.

- Disposable sterile Pipettes, only use one time.

- Glass ware and dissecting sets can be sterilized by dry heat (90 min at 160 C°).

- Media only sterilized by Filtration and should not steriled by Autoclave, because the media contain elements which are heat sensitive (glycine for example).

- Using 0.45 um filter prior 0.2 um filter.

Storage

- Culture media at 4 C°
- Fetal calve serum at – 20 C°
- Short term storage at – 80 C°
- Long term storage at – 196 C° (Liquid nitrogen)

Routine cleaning the tissue culture Lab

Daily tasks

- Mop the floor with detergent at the end of the day

Weekly tasks

- Wipe all general surfaces with 70% alcohol.
- wash L flow with detergent followed by 70% alcohol.
- replace the humidity water in the incubator.
- pour disinfectant in the sink.

Every three months

-dismantle the inside of the incubator wash with detergent and alcohol

-wash pre filter of Laminar flow with detergent rinse and replace.

Personal hygiene

-Wear a clean lab coat reserved for TC

-Disposable foot cover

-Paper hairnet

-Try not to sneeze or talk while working

-wash hands before and after work

-Fingernails should be cut

-No food or drinks

Media

- Most important Component, it provide essential elements and growth factor and hormones essential for cells growth.

- Foaming increase the contamination and limits the gaseous diffusion.

- Major problems for monoclonal production are

- Foam formation

- Damage/ losing of cells the secrete Mab

Two types of media

- **Powder**: cheaper, take up less storage space, and have longer shelf life.

Can kept 2 years

Need large volume of water to prepared, take longer to prepare.

- **Liquid media** →only kept 1 year.

- Common media are used in Hybridoma tissue culturing : RPMI-1640 and DEME

Essential additives (add to media)

- PH indicator
- Using phenol red as PH indicator.
- PH consider one of the most important factor for cell growth.
- Most normal mammalian cells growth at pH 7.4
- Optimal Temperature for most mammalian cells growth 36-37 C°

- Color of media change depend on PH

PH = 7.→purple

PH = 7.4 →red

PH= 7 →orange

PH = 6.5→yellow

- To maintain the optimal PH in culture→ buffer

a-bicarbonate buffer

b-HEPES: avoided at fusion experiment

c-Atmospheric CO_2 (5%)

d. Sodium pyruvate

- Sodium pyruvate + CO_2 · bicarbonate buffer

Glutamine

- Unstable in solution: 3 weeks at 40C and 1 week at 370C

- Purchased as 100x frozen or lyophilized powder

- Media is only sterilized by filtration and should not be autoclaved because Glutamine will be damaged, using 0.45 um filters prior to 0.2 um.

Antibiotic

- Penicillin (100ug/ml) →against gram positive Bacteria

- Streptomycin (100ug/ml)→against gram negative Bacteria

Beta-mercaptoethanol

- Enhance Ab production by spleen cells and renders hybridoma less dependent on particular patches of serum.

Serum

- Zero fetal calf serum media use for washing
- 5-10 % FCS in hybridoma culture
- 20% FCS in fusion and cloning
- 90% FCS in storage Hybridoma cells in Liquid nitrogen.

To test the Sterility of media and serum

- Small sample incubate at room temperature and other sample incubate at 37C° up to one week.

- If the contamination occur, contaminated media should autoclaved and discard, or putting contaminated culture into 2.5% hyperchlorite solution.

Some basic preventive measures may be taken:

a. Checking sterilizing procedures (autoclave and oven procedures);

b. Checking sterility of laminar flow hoods;

c. Regular checks on cultures;

d. Each worker having his or her own set of media; and

e. Disposal of contaminated cultures rather than attempting to decontamination

f. Avoid multiple sampling of stock solution to minimize contamination.

Contamination

• Most common and primary Challenge in tissue culturing is the contamination.

• Contamination divide to

- Chemical contamination
- Biological contamination

• Biological contamination→viruses, Bactria, fungi, myco plasma.

• The best and oldest way to eliminate contamination is Autoclaved and discard contaminated cultures

Effect of Biological contamination

- Compete for nutrients with host cells

- Secrete byproduct (acidic and basic byproduct) which cause change in PH that lead to ceases (stop) cells growth.

- Degrade Arginine and purine which inhibit the synthesis of histones and nucleic acids.

- H_2O_2 production which directly toxic cells.

- Most common contamination in cell culture

- Bacteria
- Yeast
- Mold

Bacteria

- Because of their ubiquity, size, and fast growth rates, bacteria, along with yeasts and molds, are the most commonly encountered biological contaminants in cell culture.

- Bacterial contamination is easily detected by visual inspection of the culture within a few days of it becoming infected; infected cultures usually appear cloudy (i.e., turbid), sometimes with a thin film on the surface.

- Sudden drops in the pH of the culture medium is also frequently encountered appeared as yellow color (phenol red).

- Under a low power microscope, the bacteria appear as tiny, moving granules between the cells, and observation under a high-power microscope can resolve the shapes of individual bacteria.

- Recommended antibiotics: Pen & strep & Gentamicin

Yeasts

- are unicellular eukaryotic microorganisms in the kingdom of Fungi, ranging in size from a few μms (typically) up to 40 μms.

- cultures contaminated with yeasts become turbid.

- There is very little change in the pH of the culture contaminated by yeasts until the contamination becomes heavy, at which stage the pH usually increases.

- Under microscopy, yeast appear as individual ovoid or spherical particles, that may bud off smaller particles

- Can be avoided by the use of amphotericin B or nystatin

Mycoplasma

- Mycoplasma are simple bacteria that lack a cell wall, and they are smallest self-replicating organism. (their extremely small size (less than one μm)

- Mycoplasma are very difficult to detect until they achieve extremely high densities and cause the cell culture to deteriorate; until then, there are often no visible signs of infection.

- The only assured way of detecting mycoplasma contamination is by testing the cultures periodically using fluorescent staining (e.g., Hoechst, ELISA, PCR).

- Treated with Tylosin

Viruses

• Small, non-enveloped viruses are the most worrisome of the viral contaminants

• Because most viruses have very stringent requirements for their host, they usually do not adversely effect cell cultures from species other than their host.

• Viral contamination events are a rare occurrence in the biopharmaceutical industry However, using virally infected cell cultures can present a serious health hazard to the laboratory personnel, especially if human or primate cells are cultured in the laboratory.

• Some viral contamination present a serious threat to TC:

a. This is firstly because of the difficulty in specifically detecting and identifying viral contaminants which often do not produce any cytopathic effect

b. And secondly because of inability to eradicate them once they have established a persistentinfection.

• These problems stem from their very small size (varying from approximately 25-500 nm in diameter), and from their existence as obligate intracellular parasites.

- While some viruses are capable of causing morphological changes to infected cells (e.g. Cytopathic effect) which are detectable by microscopy.

- Some viral contaminations result in the integration of the viral genome as provirus, this causes no visual evidence, by means of modification of the cellular morphology.

- Viral infection of cell cultures can be detected by ELISA assays

- Virus production from cell lines, are potentially dangerous for other cell cultures (in research labs) by cross contaminations, or for operators and patients (in the case of the production of injectable biologicals) because of potential infection.

Chemical contamination

- often overlooked as a cause of cell growth problems, is caused by nonliving substances that produce unwanted effects on a culture system.

- These can include impurities in media, sera, water and endotoxins, plasticizers, and fluorescent lights and incubators.